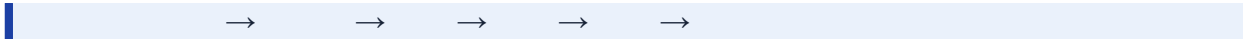


PROJECT UPGRADE PROPOSAL

I



RAG



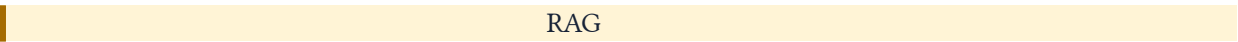
V1.0

2026 7 24



AI

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
7. Chat Agent
- 8.
9. UI
- 10.
- 11.
- 12.
- 13.
- 14.



1.1

FastAPI Vue 3 MySQL Chroma LangChain API

AI

1.2

- “ ” “ ”
- AI PPT
-
- PWA
-
- OCR
- “ ”

“ AI ”

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2.1 □ □ □ □

“ ”

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2.2 □ □ □ □ □ □

- 1.
- 2. AI
- 3.
- 4.
- 5.
- 6.
- 7. AI
- 8.

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3.1 □ □ □ □

“ ”

AI

3.2 □ □ □ □ □ □ □

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3.3

“ ” ”

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- 24 7 30
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3.4

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	API/RSS/Sitemap/ URL	
	PDF/HTML	
	URL	
	BM25/Embedding	

3.5

AI

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		66 33

3.6

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4.1 □ □ □ □

4.2 □ □ □ □ Agent □ □

- 1.
- 2. Agent
- 3.
- 4.
- 5.
- 6. PPT
- 7.
- 8. Word PPT Markdown

4.3 □ □ □ □

□ □	□ □ □ □	□ □ □ □
		/
PPT		/ / Agent
		/
	AI	

4.4 PPT □ □ □ □

- PPTJSON
- .pptx
- -
 -
 - Logo
 -

4.5 □□□□□□□□

Embedding

PPT

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5.1 □□□□

□□	□□□□	□□□□□
		AI
		/
		AI
		AI

5.2 □□□□□



5.3 □□□□□□

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AI

5.4 □□□□

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6.1 □□□□□□□□

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		/
		/

6.2 □□□□□□□

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- /

6.3 □□□□□□

PWA

“

”

□□Chat □ Agent □□□

7.1 □□□□

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Chat		
Agent	PPT	

7.2 Agent □□□□

- Agent
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I

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7.3 □□□□

AI

Agent

“ ” “ ”

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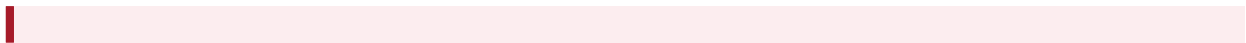
8.1 □□□□

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-
- OCR
-
-

8.2 □□□□

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8.3 □□□□□



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-
- PPT
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□□□□□□□ UI □□

9.1 □□□□

AI

9.2 □□□□□□

□□	□□□□	□□□□□

9.3 □□□□□□

- “ ”
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- “ ”
- AI AI
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9.4 □□□□

- “ AI ”
- “AI”
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10.1 □□□□

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	Vue 3 PWA	
API	FastAPI JWT	
Agent		
	RAG Embedding Chroma/ Milvus	
	Redis Worker	PPT
	MySQL OSS	PPT
	Embedding API	

10.2

SourceRegistry		domain authority_level adapter frequency enabled
DiscoveryQuery		course_id chapter_id keywords time_range
DiscoveryJob		status progress started_at error_summary
MaterialCandidate		url publisher published_date relevance novelty
PolicyChange		old_source new_source change_type evidence
AgentRun/AgentStep		user_id context status input output model
GeneratedArtifact	PPT	artifact_type content_json version source_ids
ArtifactTemplate		owner_type theme layout_config asset_refs
CommunityPost/Reply		class_id chapter_id author_id visibility status
LearningCard		chapter_id citation understanding case review_status
Notification		audience level deep_link read_time dedupe_key
MediaAsset/Transcript		object_key media_type retention transcript permission

10.3 API

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- Agent
-
-
- OCR

10.4

PPT Embedding

11.1

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/PPT			

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/AI			

11.2 □□□□□□□

- AI
- PDF
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- AI
- AI

11.3 □□□□□

- Agent
- AI
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12.1 □□□□□

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0		/ /	1—2
1			2—3
2	PPT	Agent	3—4
3		AI	2—3
4	PWA		2
5		OCR	2—3

12.2 □□□□□□□

- 1.
2. Agent PPT

3. AI

“ ”

□ □ □ □ □ □ □ □ □ □ □ □ □ □

13.1 □ □ □ □ □

2 vCPU 2GB FastAPI MySQL Nginx Agent PPT 50

13.2 □ □ □ □ □ □ □

□ □	□ □ □ □	□ □
ECS	4 vCPU / 8GB	GPU
	10Mbps	50 PDF
	40—50GB ESSD	
	MySQL	RDS
	Redis 1 Worker	PPT Embedding
	OSS	PPT
API	2 Uvicorn Worker	Worker

13.3 □ □ □ □ □ □

50 AI 8 vCPU 16GB MySQL RDS Redis AI
API CPU

□ □ □ □ □ □ □ □ □ □ □ □ □ □

14.1 □ □ □ □ □ □ □

□ □	□ □ □ □	□ □ □ □
		≥95% ≤5%
		100%
		≥95%
	OCR/	
	50 API	OOM AI
		≤5 ≤2

14.2 □ □ □ □ □ □ □

□ □	□ □	□ □ □ □
AI		
	OOM	4 8GB OSS Redis

14.3 ☐ ☐ ☐ ☐ ☐ ☐

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|----|----------------|-----|--------|-------|---------|
| 1. | main | | | | Alembic |
| 2. | | | | | |
| 3. | SourceRegistry | | | | 3—5 |
| 4. | Agent | → | → | →PPT | → |
| 5. | | | | | |
| 6. | 4 | 8GB | 10Mbps | Redis | OSS |
| 7. | | | | | |

“ ”

A

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- AI
- GPU

□ □ **B** □ □ □ □ □ □ □ □ □

□ □	□ □ □ □ □
	AI
	PPT
PPT	
	AI
	PWA
Agent	
	AI